

Rule	Original Diagram	Equivalent Diagram
1. Blocks in cascade		
2. Moving a summing point ahead of a block		
3. Moving a summing point beyond a block		
4. Moving a take-off point ahead of a block		
5. Moving a take-off point beyond a block		
6. Rearrangement of summing point		
7. Eliminating a forward loop		
8. Eliminating feedback loop		

1.24. SIGNAL FLOW GRAPH

The process of block diagram reduction technique is time consuming because at every stage modified block diagram is to be redrawn. A simple method was developed by S.J. Mason which is known as signal flow graph. This method is very simple and does not require any reduction technique. Signal flow graph is applicable to the linear systems.

A signal flow graph is a diagram which represents a set of simultaneous equations.

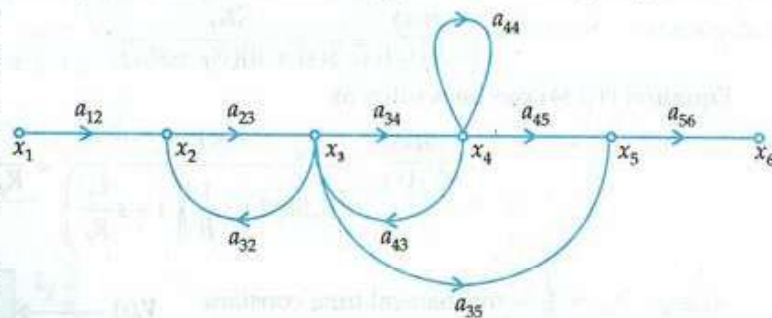


Fig. 1.101

Signal flow graph consists of nodes and these nodes are connected by a directed line called branches. Every branch of signal flow graph having an arrow, which represents the flow of signal.

The following terms are associated with the signal flow graph.

1. **Input node or source node** : An input node is a node which has only outgoing branches. For example x_1 is the input node.
2. **Output node or sink node** : An output node is a node that has only one or more incoming branches. e.g. x_6 is the output node.
3. **Mixed nodes** : A node having incoming and outgoing branches is known as mixed nodes. For example x_2, x_3, x_4 and x_5 are the mixed nodes.
4. **Transmittance** : Transmittance also known as transfer function, which is normally written on the branch near the arrow. For example a_{12}, a_{23} etc.
5. **Forward path** : Forward path is a path which originates from the input node and terminates at the output node and along which no node is traversed more than once.

For example in Fig 1.101 there are two forward paths.

1. x_1 to x_2 to x_3 to x_4 to x_5 to x_6
2. x_1 to x_2 to x_3 to x_5 to x_6

6. **Loop** : Loop is a path that originates and terminates on the same node and along which no other node is traversed more than once.

For example x_2 to x_3 to x_2
 x_3 to x_4 to x_3

7. **Self loop** : It is a path which originates and terminates on the same node. For example x_4 to x_4
8. **Path gain** : The product of the branch gains along the path is called path gain. For example the gain of the path x_1 to x_2 to x_3 to x_4 to x_5 to x_6 is $a_{12} a_{23} a_{34} a_{45} a_{56}$
9. **Loop gain** : The gain of the loop is known as loop gain. For example the gain of the loop x_2 to x_3 to x_2 is $a_{23} a_{32}$.
10. **Non-touching loops** : Non touching loops having no common nodes branch and paths. For example the loops x_2 to x_3 to x_2 , and x_4 to x_4 are non-touching loops.

1.25. PROPERTIES OF SIGNAL FLOW GRAPH

1. Signal flow graph is applicable to linear time-invariant systems.
2. The signal flow is only along the direction of arrows.
3. The value of variable at each node is equal to the algebraic sum of all signals entering at that node.
4. The gain of signal flow graph is given by Mason's formula.
5. The signal gets multiplied by the branch gain when it travels along it.
6. The signal flow graph is not be the unique property of the system.

1.26. COMPARISON OF BLOCK DIAGRAM AND SIGNAL FLOW GRAPH METHOD

S.No.	Block Diagram	SFG
1.	Applicable to linear time invariant systems only.	Applicable to linear time invariant system.
2.	Each element is represented by block.	Each variable is represented by node.
3.	Summing point and take off points are separate.	Summing and take off points are absent.
4.	Self loop do not exist.	Self loop can be exist.
5.	It is time consuming method.	Require less time by using Mason gain formula.
6.	Block diagram is required at each & every step.	At each step it is not necessary to draw SFG.
7.	Transfer function of the element is shown inside the corresponding block.	Transfer function is shown along the branches connecting the nodes.
8.	Feedback path is present.	Feedback loops are used.

1.27. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM EQUATIONS

Consider the following sets of equations

$$\begin{aligned}
 y_2 &= t_{21} y_1 + t_{23} y_3 \\
 y_3 &= t_{32} y_2 + t_{33} y_3 + t_{31} y_1 \\
 y_4 &= t_{43} y_3 + t_{42} y_2 \\
 y_5 &= t_{54} y_4 \\
 y_6 &= t_{65} y_5 + t_{64} y_4
 \end{aligned}$$

where y_1 is the input and y_6 is the output.

First of all draw the nodes. In the given example there are six nodes. From the first equation it is clear that the y_2 is the sum of two signals. Similarly, y_3 is the sum of three signals and so on. Insert the branches with proper transmittance to connect the nodes.

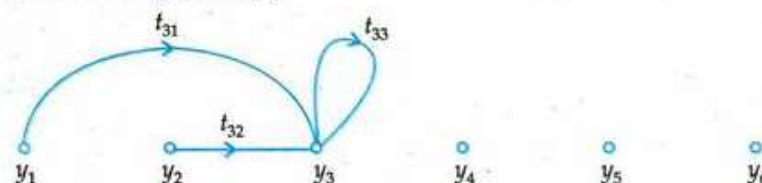
Step 1 : Draw the nodes



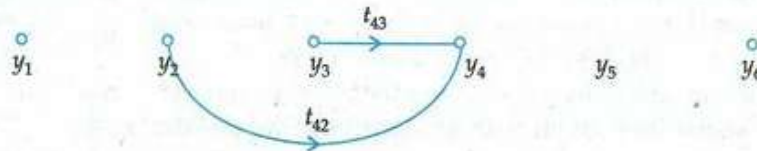
Step 2 : Draw the SFG for equation (1)



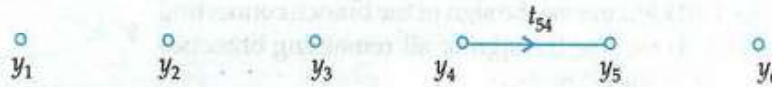
Step 3 : Draw the SFG for equation (2)



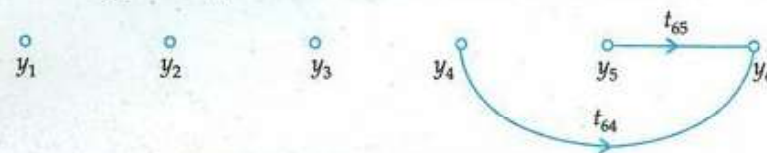
Step 4 : Draw SFG for equation (3)



Step 5 : Draw SFG for equation (4)



Step 6 : Draw SFG for equation (5)



Step 7 : Draw the complete signal flow graph with the help of above graphs.

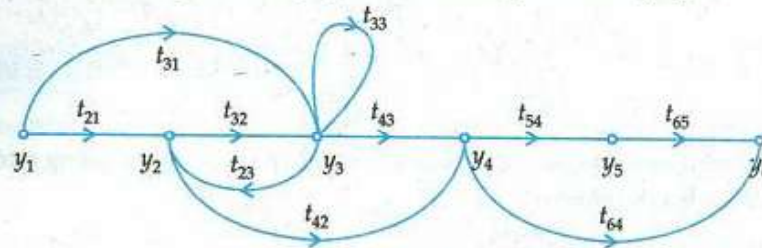


Fig. 1.102.

1.28. SIGNAL FLOW GRAPH FOR DIFFERENTIAL EQUATIONS

Consider the following differential equation

$$y''' + 3y'' + 5y' + 2y = x \quad \dots(1.135)$$

Step 1 : Solve the eqn 1.135 for the highest order

$$y''' = x - 3y'' - 5y' - 2y$$

Step 2 : Consider the left hand term (highest order derivative) as dependent variable and all other terms on right hand side as independent variables.

construct the branches of signal flow graph as shown in Fig. (1.103(a)).

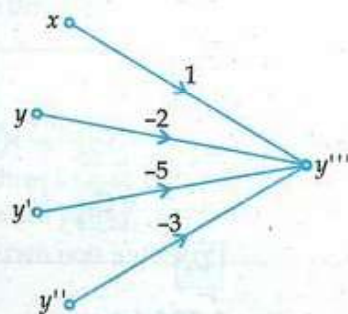


Fig. 1.103 (a)

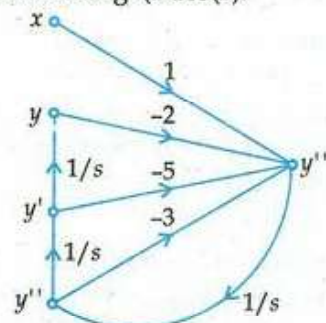


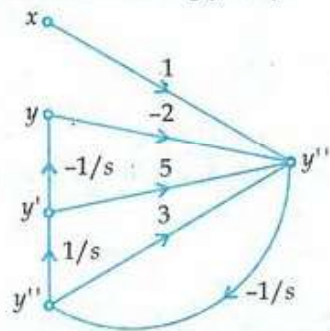
Fig. 1.103 (b)

Step 3 : Connect the nodes of highest order derivative to the node whose order is lower than this and so on. The flow of the signal will be from higher node to the lower order node and transmittance will be $1/s$ as shown in Fig. 1.103 (b).

Step 4 : Reverse the sign of a branch connecting the p^{th} node to the q^{th} node of a signal flow graph without disturbing the transfer function.

Consider the Fig. 1.103(b), reverse the sign of the branch connecting y''' to y'' , it is necessary to reverse the sign of all remaining branches entering as well as leaving the q^{th} node.

Similarly, reverse the sign of branch connecting y'' to y' .



By reversing the sign, we have already reverse the sign of branch connecting y' to y and therefore further reversal of sign is not required.

Step 5 : Redraw the signal flow graph (SFG).

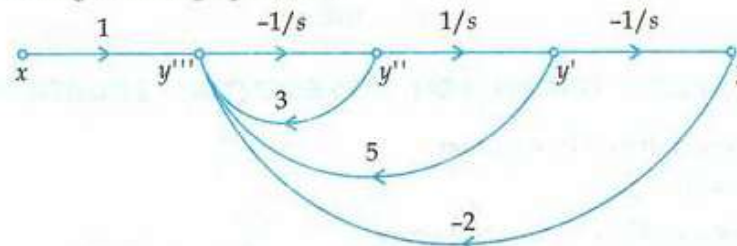


Fig. 1.103 (c)

1.29. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM BLOCK DIAGRAM

Rules 1. All variables, summing points and take off points are represented by nodes.

2. If a summing point is placed before a take off point in the direction of signal flow, in such case represent the summing point and takeoff point by a single node.
3. If a summing point is placed after a takeoff point in the direction of signal flow, in such case,

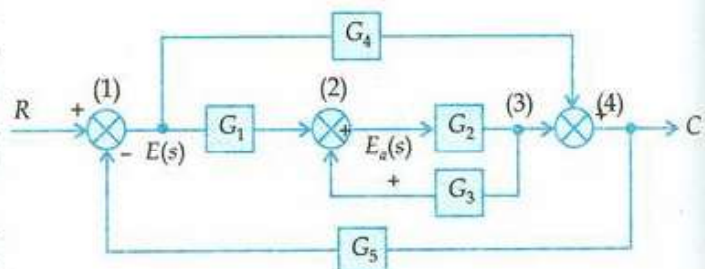


Fig. 1.104 (a)

represent the summing point and takeoff point by separate nodes connected by a branch having transmittance unity.

Consider the block diagram shown in Fig 1.104(a), the corresponding SFG is shown in Fig. 1.104 (b).

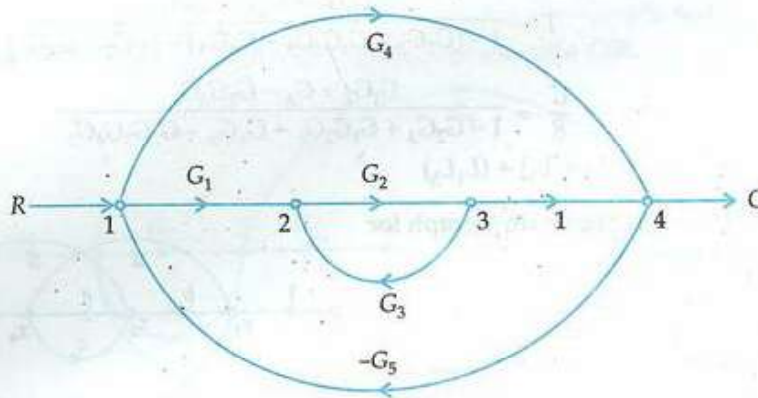


Fig. 1.104 (b)

1.30. MASON'S GAIN FORMULA

The overall transmittance or graph transmittance between the source node and sink node is given by Mason's gain formula. Mason's gain formula for signal flow graph is as follows .

$$T = \frac{\sum g_k \Delta_k}{\Delta} \quad \dots(1.136)$$

where T = transfer function

$\Delta = 1 - [\text{sum of all individual loop gain}] + [\text{sum of all possible gain products of two non-touching loops}] - [\text{sum of all possible gain products of three non-touching loops}] + \dots$

g_k = gain of the k^{th} forward path

Δ_k = the part of Δ not touching the k^{th} forward path.

Δ_1 - the part of Δ not touching the 1st forward path

Consider the signal flow graph shown in Fig. (1.104(b)). There are two forward paths (i) 1-2-3-4 (ii) 1-4 therefore gain of the two paths will be

$$g_1 = G_1 G_2$$

$$g_2 = G_4$$

There are three individual loops

$$L_1 = G_2 G_3 \quad (2-3-2)$$

$$L_2 = -G_1 G_2 G_5 \quad (1-2-3-4-1)$$

$$L_3 = -G_4 G_5 \quad (1-4-1)$$

Since all three loops touching the forward path g_1 , therefore $\Delta_1 = 1 - 0 = 1$ The first loop L_1 do not touch the forward path g_2 , therefore $\Delta_2 = 1 - G_2 G_3$

There are two non touching loops L_1 and L_3

$$\Theta \quad L_1 L_3 = -G_2 G_3 G_4 G_5$$

Apply Mason's gain formula

$$T = \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$T = \frac{G_1 G_2 + G_4 (1 - G_2 G_3)}{1 - (G_2 G_3 - G_1 G_2 G_5 - G_4 G_5) - (-G_2 G_3 G_4 G_5)}$$

$$\frac{C}{R} = \frac{G_1 G_2 + G_4 - G_2 G_3 G_4}{1 - G_2 G_3 + G_1 G_2 G_5 + G_4 G_5 - G_2 G_3 G_4 G_5}$$

Note : where $\Delta = 1 - (L_1 + L_2 + L_3) + (L_1 L_3)$

EXAMPLE 1.40. Draw the signal flow graph for the following set of equations.

$$x_2 = x_1 + ax_5$$

$$x_3 = bx_2 + cx_4$$

$$x_4 = dx_2 + ex_3$$

$$x_5 = fx_4 + gx_3$$

$$x_6 = x_5$$

Solution : The required signal flow graph is shown in Fig. 1.105.

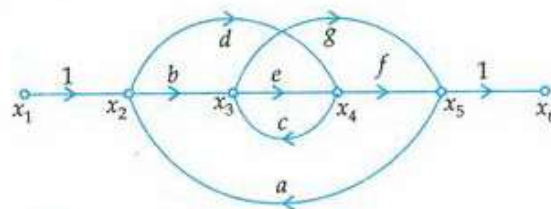


Fig. 1.105.

EXAMPLE 1.41. For the system represented by the given equations find the transfer function x_5/x_1 by the help of signal flow graph technique.

$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$

$$x_3 = a_{23}x_2$$

$$x_4 = a_{34}x_3 + a_{44}x_4$$

$$x_5 = a_{35}x_3 + a_{45}x_4$$

where x_1 is the input variable and x_5 is the output variable.

(R.M.L. University Faizabad, 2002)

Solution : There are two forward paths: (1) x_1 to x_2 to x_3 to x_4 to x_5

(2) x_1 to x_2 to x_3 to x_5

The gain of the paths $g_1 = a_{12} a_{23} a_{34} a_{45}$

$$g_2 = a_{12} a_{23} a_{35}$$

Gain of individual loops

$$L_1 = a_{23} a_{32}$$

$$L_2 = a_{23} a_{34} a_{42}$$

$$L_3 = a_{23} a_{34} a_{45} a_{52}$$

$$L_4 = a_{23} a_{35} a_{42}$$

$$L_5 = a_{44}$$

Gain of two nontouching loops

$$L_1 L_5 = a_{23} a_{32} a_{44}$$

$$L_4 L_5 = a_{23} a_{35} a_{42} a_{44}$$

Since, all the loops touches the forward path (1) $\therefore \Delta_1 = 1 - 0 = 1$

loop L_5 do not touch the second forward path $\therefore \Delta_2 = 1 - a_{44}$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + (L_1 L_5 + L_4 L_5)$$

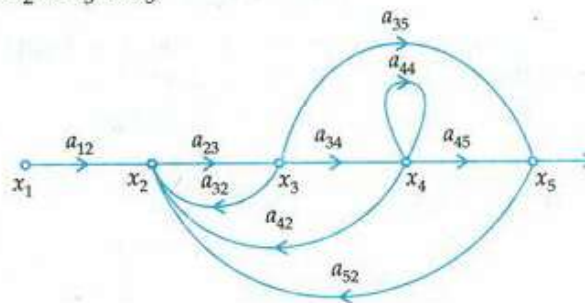


Fig. 1.106.

$$\therefore \frac{x_5}{x_1} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$\frac{x_5}{x_1} = \frac{a_{12}a_{23}a_{34}a_{45} + a_{12}a_{23}a_{35}(1-a_{44})}{1 - (a_{23}a_{32} + a_{23}a_{34}a_{42} + a_{23}a_{34}a_{45}a_{52} + a_{23}a_{35}a_{52} + a_{44}) + (a_{23}a_{32}a_{44} + a_{23}a_{35}a_{52}a_{44})}$$

EXAMPLE 1.42. For the given signal flow graph find the ratio C/R .

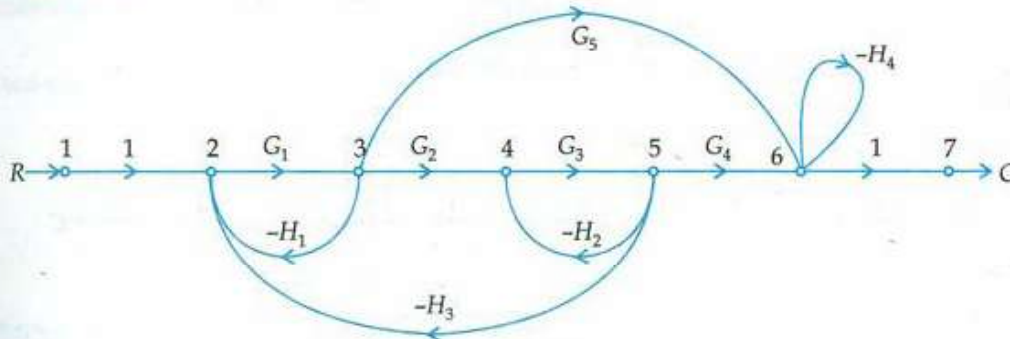


Fig. 1.107.

The gain of the forward paths $g_1 = G_1 G_2 G_3 G_4$

$$g_2 = G_1 G_5$$

Individual loops

$$L_1 = -G_1 H_1$$

$$L_2 = -G_3 H_2$$

$$L_3 = -G_1 G_2 G_3 H_3$$

$$L_4 = -H_4$$

Two non-touching loop

$$L_1 L_2 = G_1 H_1 G_3 H_2$$

$$L_1 L_4 = G_1 H_1 H_4$$

$$L_2 L_4 = G_3 H_2 H_4$$

$$L_3 L_4 = G_1 G_2 G_3 H_3 H_4$$

Three non-touching loops $L_1 L_2 L_4 = -G_1 H_1 G_3 H_2 H_4$

$$\Delta_1 = 1 - 0$$

$$\Delta_2 = 1 + G_3 H_2$$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1 L_2 + L_1 L_4 + L_2 L_4 + L_3 L_4) - (L_1 L_2 L_4)$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3 G_4 + G_1 G_5 (1 + G_3 H_2)}{1 + G_1 H_1 + G_3 H_2 + G_1 G_2 G_3 H_3 + H_4 + G_1 G_3 H_1 H_2 + G_1 H_1 H_4 + G_3 H_2 H_4 + G_1 G_2 G_3 H_3 H_4 + G_1 G_3 H_1 H_2 H_4}$$

EXAMPLE 1.43. Obtain signal flow graph representation for a system whose block diagram is given below and using Mason's gain formula determine the ratio $\frac{C}{R}$.

(R.M.L. Univ. Faizabad 2001; Linear System Theory)

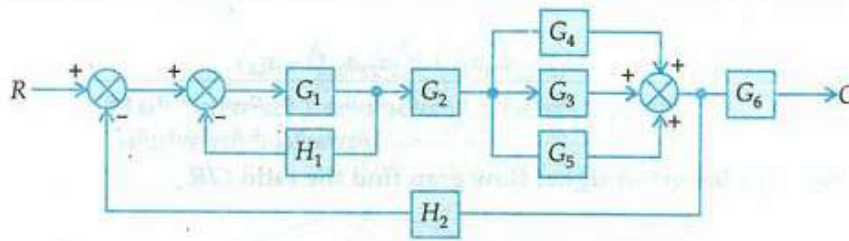


Fig. 1.108 (a).

Solution :

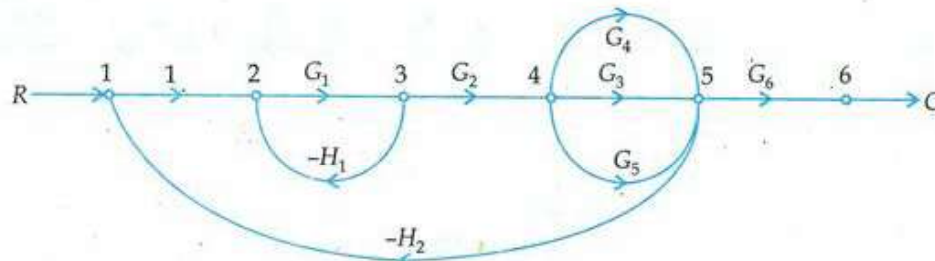


Fig. 1.108 (b)

The signal flow graph is shown in Fig. 1.108(a). Consider the SFG

$$g_1 = G_1 G_2 G_3 G_6$$

$$\Delta_1 = 1 - 0 = 1$$

$$g_2 = G_1 G_2 G_4 G_6$$

$$\Delta_2 = 1 - 0 = 1$$

$$g_3 = G_1 G_2 G_5 G_6$$

$$\Delta_3 = 1 - 0 = 1$$

Individual loops

$$L_1 = -G_1 H_1$$

$$L_2 = -G_1 G_2 G_3 H_2$$

$$L_3 = -G_1 G_2 G_4 H_2$$

$$L_4 = -G_1 G_2 G_5 H_2$$

Two non touching loops = None

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4)$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 G_6 + G_1 G_2 G_4 G_6 + G_1 G_2 G_5 G_6}{1 + G_1 H_1 + G_1 G_2 G_3 H_2 + G_1 G_2 G_4 H_2 + G_1 G_2 G_5 H_2}$$

EXAMPLE 1.44. Obtain the Transfer function for $\frac{C(s)}{R(s)}$ given block diagram using block diagram reduction technique or signal flow graph (Mason's gain formula).

(R.M.L. Univ. Faizabad 2001; Control System)

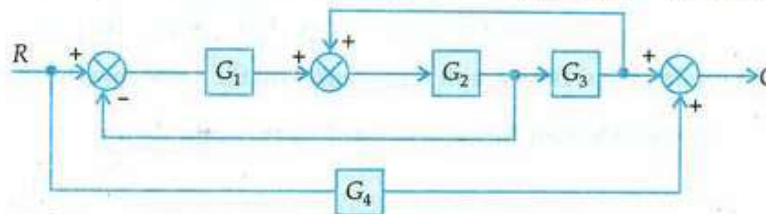


Fig. 1.109 (a).

Solution : By signal flow graph SFG is shown in fig 1.109a.

Gain of forward paths : $g_1 = G_1 G_2 G_3$

$$g_2 = G_4$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 + G_1 G_2 - G_2 G_3$$

Individual loops :

$$L_1 = -G_1 G_2$$

$$L_2 = G_2 G_3$$

Two non-touching loops : None

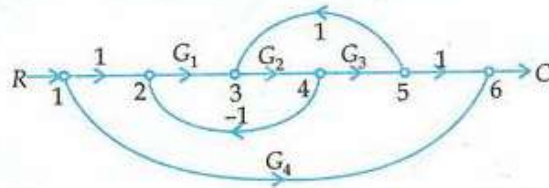


Fig. 1.109 (b)

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3 + G_4 (1 + G_1 G_2 - G_2 G_3)}{1 + G_1 G_2 - G_2 G_3}$$

EXAMPLE 1.45. Draw the signal flow graph and determine $\frac{C}{R}$ for the block diag. shown in

Fig. 1.110.

(KNIT, Sultanpur, 1999-2000)

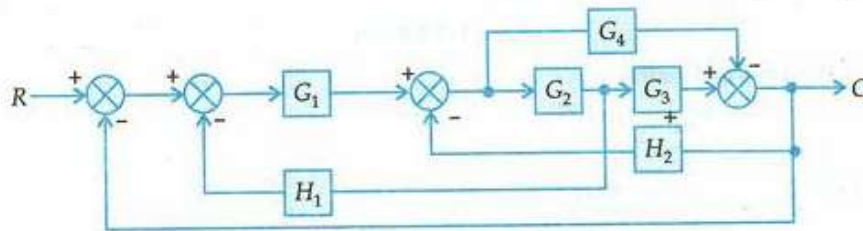


Fig. 1.110 (a).

Solution : The SFG is shown in Fig. 1.110 (a)

The gain of the forward paths

$$g_1 = G_1 G_2 G_3$$

$$\Delta_1 = 1$$

$$g_2 = -G_1 G_4$$

$$\Delta_2 = 1$$

Individual loops

$$L_1 = -G_1 G_2 H_1$$

$$L_2 = -G_2 G_3 H_2$$

$$L_3 = -G_1 G_2 G_3$$

$$L_4 = G_1 G_4$$

$$L_5 = G_4 H_2$$

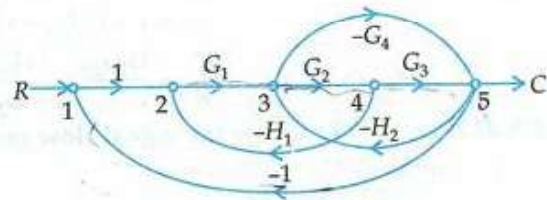


Fig. 1.110 (b)

$$\frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3 - G_1 G_4}{1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 - G_1 G_4 - G_4 H_2}$$

EXAMPLE 1.46. Obtain the transfer function C/R of the block diagram shown in Fig. 1.111.

(KNIT, Sultanpur)

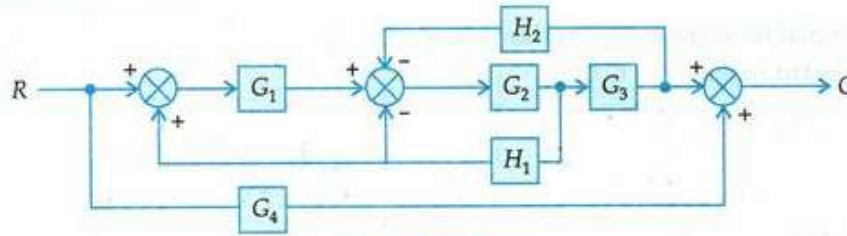


Fig. 1.111 (a).

Solution :

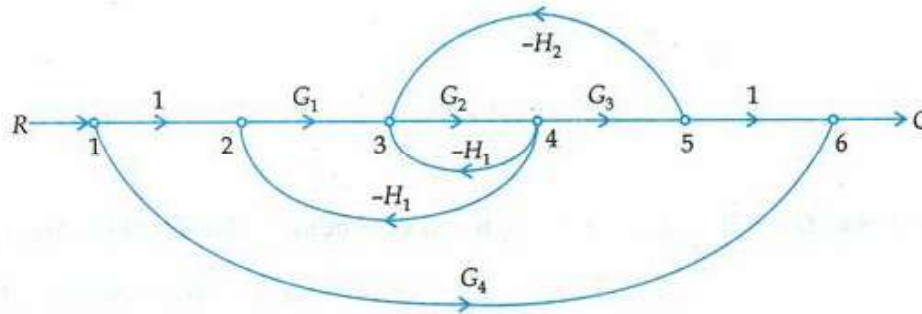


Fig. 1.111 (b)

The gain of the forward paths

$$g_1 = G_1 G_2 G_3$$

$$g_2 = G_4$$

Individual loops

$$L_1 = -G_2 H_1$$

$$L_2 = -G_1 G_2 H_1$$

$$L_3 = -G_2 G_3 H_2$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 - (-G_2 H_1 - G_1 G_2 H_1 - G_2 G_3 H_2)$$

$$\Delta_2 = 1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2$$

$$\Delta = 1 - (L_1 + L_2 + L_3)$$

$$\Delta = 1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2)}{1 + G_2 H_1 + G_1 G_2 H_1 + G_2 G_3 H_2}$$

EXAMPLE 1.47. Draw the signal flow graph for the network shown in Fig. 1.112, take V_3 as output node.

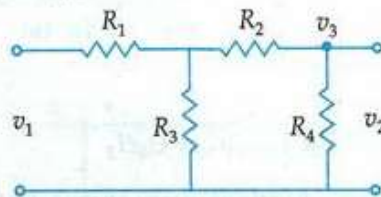


Fig. 1.112 (a)

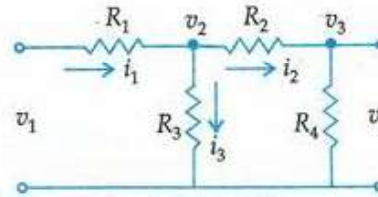


Fig. 1.112 (b)

Solution : From 1.112a : $i_1 = \frac{V_1}{R_1} - \frac{V_2}{R_1}$

$$V_2 = i_3 R_3 = R_3 (i_1 - i_2) = R_3 i_1 - R_3 i_2$$

$$i_2 = \frac{V_2}{R_2} - \frac{V_3}{R_2}$$

$$V_3 = R_4 i_2$$

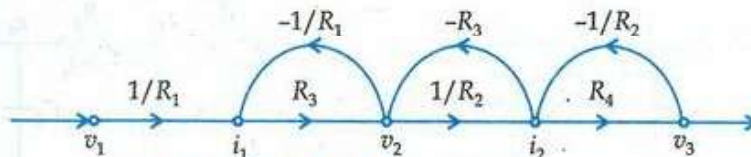


Fig. 1.112 (c) Signal flow graph.

EXAMPLE 1.48. Obtain the transfer function C/R from the signal flow graph shown in Fig. 1.113.

Solution : The gain of the forward paths

$$g_1 = G_2 G_4 G_6$$

$$g_2 = G_3 G_5 G_7$$

$$g_3 = G_2 G_1 G_7$$

$$g_4 = G_3 G_8 G_6$$

$$g_5 = -G_2 G_1 H_2 G_8 G_6$$

$$g_6 = -G_3 G_8 H_1 G_1 G_7$$

Individual loops :

$$L_1 = -G_4 H_1$$

$$L_2 = -G_5 H_2$$

$$L_3 = H_1 G_1 H_2 G_8$$

Two non touching loops :

$$L_1 L_2 = G_4 H_1 G_5 H_2$$

$$\Delta_1 = 1 + G_5 H_2$$

$$\Delta_2 = 1 + G_4 H_1$$

$$\Delta_3 = 1$$

$$\Delta_4 = 1$$

$$\Delta_5 = 1$$

$$\Delta_6 = 1$$

$$\Delta = 1 + G_4 H_1 + G_5 H_2 - H_1 G_1 H_2 G_8 + G_4 H_1 G_5 H_2$$

$$\therefore \frac{C}{R} = \frac{g_1 \Delta_1 + g_2 \Delta_2 + g_3 \Delta_3 + g_4 \Delta_4 + g_5 \Delta_5 + g_6 \Delta_6}{\Delta}$$

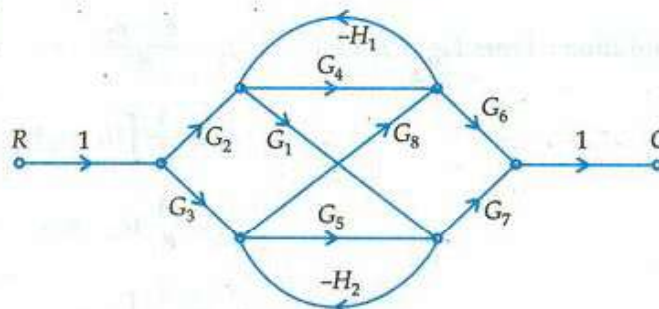


Fig. 1.113.

$$\frac{C}{R} = \frac{G_2 G_4 G_6 (1 + G_5 H_2) + G_3 G_5 G_7 (1 + G_4 H_1) + G_1 G_2 G_7 + G_3 G_8 G_6 - G_2 G_1 H_2 G_8 G_6 - G_3 G_8 H_1 G_1 G_7}{1 + G_4 H_1 + G_5 H_2 - H_1 G_1 H_2 G_8 + G_4 H_1 G_5 H_2}$$

Ans.

EXAMPLE 1.49. Draw the block diagram, and signal flow graph and find out the transfer function of the circuit shown in Fig. 1.114 (a).

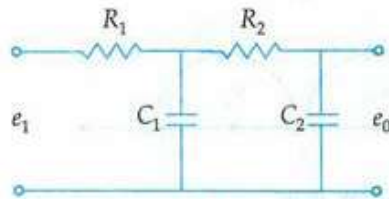


Fig. 1.114 (a)

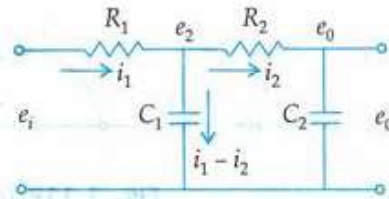


Fig. 1.114 (b)

Solution : From Fig. 1.114 (b): $i_1 = \frac{e_i - e_2}{R_1}$

$$e_2 = \frac{1}{C_1} \int (i_1 - i_2) dt$$

$$i_2 = \frac{1}{R_2} (e_2 - e_0)$$

$$e_0 = \frac{1}{C_2} \int i_2 dt$$

Take Laplace transform of above equations

$$I_1(s) = \frac{1}{R_1} [E_i(s) - E_2(s)]$$

$$E_2(s) = \frac{1}{sC_1} [I_1(s) - I_2(s)]$$

$$I_2(s) = \frac{1}{R_2} [E_2(s) - E_0(s)]$$

$$E_0(s) = \frac{1}{sC_2} [I_2(s)]$$

On the basis of above equation draw the block diagram

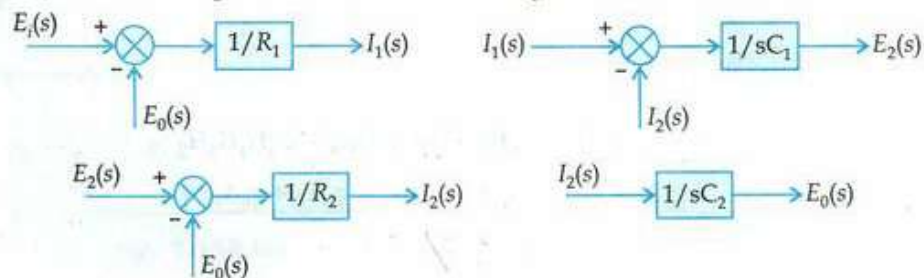


Fig. 1.114 (c). Block diagram

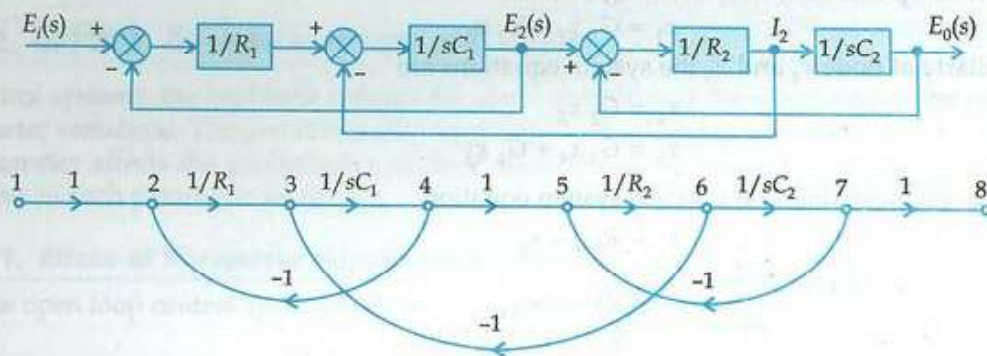


Fig. 1.114 (d). Signal flow graph

$$g_1 = \frac{1}{R_1 R_2 C_1 C_2 s^2}$$

$$\Delta_1 = 1$$

$$L_1 = -\frac{1}{R_1 C_1 s}$$

$$L_2 = -\frac{1}{R_2 C_1 s}$$

$$L_3 = -\frac{1}{R_2 C_2 s}$$

Two non-touching loops

$$L_1 L_3 = \frac{1}{s^2 R_1 R_2 C_1 C_2}$$

$$\frac{C}{R} = \frac{\frac{1}{R_1 R_2 C_1 C_2 s^2}}{1 + \frac{1}{R_1 C_1 s} + \frac{1}{R_2 C_1 s} + \frac{1}{R_2 C_2 s} + \frac{1}{s^2 R_1 R_2 C_1 C_2}}$$

$$\frac{C}{R} = \frac{1}{1 + R_1 C_1 s + R_1 C_2 s + R_2 C_2 s + s^2 R_1 R_2 C_1 C_2}$$

Ans.

1.31. BLOCK DIAGRAM FROM SIGNAL FLOW GRAPH

Step 1: For given signal flow graph, write the system equations.

Step 2: At each node consider the incoming branches only.

Step 3: Add all incoming signals algebraically at a node.

Step 4: For + or - sign in system equations use a summing point

Step 5: For the gain of each branch of signal flow graph draw the block having the same transfer function as the gain of the branch.

Consider the following examples :

EXAMPLE 1.50. Draw the block diagram from the given signal flow graph.

Solution : The system equations are

At node x_1 , the incoming branches are from $R(s)$ and x_5

$$x_1 = 1 R(s) - 1 \cdot x_5$$

At node x_2 , there are two incoming branches

$$x_2 = 1 \cdot x_1 - H_1 x_4$$

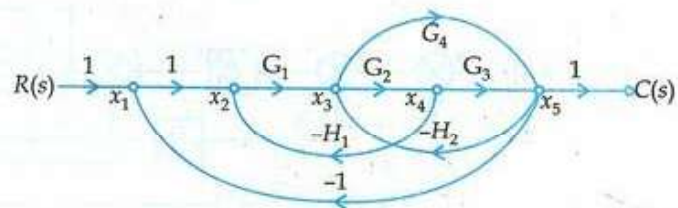


Fig. 1.115 (a)

CHAPTER 3

Error Analysis

Inside this Chapter

3.1. Classification of Control System; 3.2. Steady-State Error; 3.3. Static Error Coefficients; 3.4. Steady-State Error for Different Type of Systems; 3.5. Dynamic Error Coefficients

3.1. CLASSIFICATION OF CONTROL SYSTEM

Consider the open loop transfer function

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{s^m(1+sT_a)(1+sT_b)+\dots} \quad \dots(3.1)$$

In equation (3.1), the poles are at $s = -\frac{1}{T_a}$, $s = -\frac{1}{T_b}$... and zeros are at $s = -1/T_1$, $s = -1/T_2$... The equation having a term s^m in denominator, 'm' is the number of poles at the origin.

A system having no pole at origin of the 's' plane, is said to be type '0' (zero) system i.e., $m = 0$

If $m = 1$ i.e., 's', it means the system has a pole at origin of the s-plane and is said to be type '1' (one) system.

A system is called type '2' system if $m = 2$. and so on.

3.2. STEADY-STATE ERROR

The steady-state error is the difference between the input and output of the system during steady state. For accuracy the steady state error should be minimum.

Consider a closed loop control system shown in Fig. 3.1

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\text{or} \quad E(s) = \frac{R(s)}{1+G(s)H(s)} \quad \dots(3.2)$$

The steady state error of the system is obtained by applying final value theorem.

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot E(s) \quad \dots(3.3)$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)} \quad \dots(3.4)$$

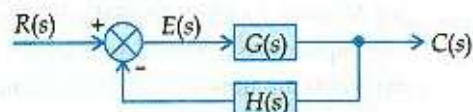


Fig. 3.1.

For unity feedback system $H(s) = 1$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)} \quad \dots(3.5)$$

From the equation (3.4) or equation (3.5) it is clear that the steady state error depends on the input and open loop transfer function.

3.3. STATIC ERROR COEFFICIENTS

(a) Static-Position Error Constant (or Coefficient) K_p

The steady state error is given by equation (3.4)

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$$

For unit step input $R(s) = \frac{1}{s}$, the steady state error is given by

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{1}{1+G(s)H(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)H(s)}$$

$$e_{ss} = \frac{1}{1+K_p} \quad \dots(3.6)$$

K_p = static position error constant = $\lim_{s \rightarrow 0} G(s)H(s)$

(b) Static Velocity Error Constant (or Coefficient) K_v

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)}$$

Steady state error with a unit ramp input is given by $[R(s) = 1/s^2]$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s G(s)H(s)}$$

$$e_{ss} = \frac{1}{K_v} \quad \dots(3.7)$$

where $K_v = \lim_{s \rightarrow 0} s G(s)H(s)$ static velocity error coefficient.

(c) Static Acceleration Error Constant K_a

The steady-state error of the system with unit parabolic input is given by

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^3} \cdot \frac{1}{1+G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)H(s)} = \frac{1}{K_a} \quad \dots(3.8)$$

where $K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$ = static acceleration constant.

3.4. STEADY-STATE ERROR FOR DIFFERENT TYPE OF SYSTEMS

1. (a) Type Zero System with Unit Step Input

$$R(s) = \frac{1}{s}$$

From equation (3.1)

$$G(s) H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots} \quad \dots(3.9)$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots} = K$$

From equation (3.6)

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+K} \quad \boxed{e_{ss} = \frac{1}{1+K}}$$

Hence, for type zero system the static position error constant K_p is finite.

(b) Type '0' System with Unit Ramp Input

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots} = 0$$

$$\therefore e_{ss} = \frac{1}{K_v} = \infty \quad \boxed{e_{ss} = \infty}$$

(c) Type '0' System with Unit Parabolic Input

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots}$$

$$K_a = 0$$

$$\therefore e_{ss} = \frac{1}{K_a} = \infty \quad \boxed{e_{ss} = \infty}$$

For type '0' system, the steady state error is infinite for ramp and parabolic inputs. Hence, the ramp and parabolic inputs are not acceptable.

2(a) Type '1' System with Unit Step Input ($m = 1$)

$$G(s) H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{s(1+sT_a)(1+sT_b)\dots}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K(1+sT_1)(1+sT_2)\dots}{s(1+sT_a)(1+sT_b)\dots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0 \quad \boxed{e_{ss} = 0}$$

(b) Type '1' System with Unit Ramp Input

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{s(1+sT_a)(1+sT_b)\dots}$$

$$K_v = K$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K} \quad \boxed{e_{ss} = \frac{1}{K}}$$

(c) Type '1' System with Unit Parabolic Input

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \frac{K(1+sT_1)(1+sT_2)\dots}{s(1+sT_a)(1+sT_b)\dots} = 0$$

$$\therefore e_{ss} = \frac{1}{K_a} = \infty \quad \boxed{e_{ss} = \infty}$$

Hence, from above relations for type '1' system, it is clear that for type '1' system step input and ramp inputs are acceptable and parabolic input is not acceptable.

3 (a) Type '2' System with Unit Step Input

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0 \quad \boxed{e_{ss} = 0}$$

(b) Type '2' System with Unit Ramp Input

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG(s)H(s) \\ &= \lim_{s \rightarrow 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots} \\ &= \infty \end{aligned}$$

$$\therefore e_{ss} = \frac{1}{K_v} = 0 \quad \boxed{e_{ss} = 0}$$

(c) Type '2' System with Unit Parabolic Input

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} s^2 G(s)H(s) \\ &= \lim_{s \rightarrow 0} s^2 \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots} = K \end{aligned}$$

$$\therefore e_{ss} = \frac{1}{K_a} = \frac{1}{K} \quad e_{ss} = \frac{1}{K}$$

Hence, for type '2' system all three inputs (step, ramp and parabolic) are acceptable. From Table 3.1, the diagonal elements are the finite values of steady state error.

Table 3.1.

	Type '0' System	Type '1' System	Type '2' System
Unit step input	$\frac{1}{1+k}$	0	0
Unit ramp input	∞	$\frac{1}{k}$	0
Unit parabolic input	∞	∞	$\frac{1}{k}$

EXAMPLE 3.1. The open loop transfer function of unity feedback system is given by

$$G(s) = \frac{50}{(1+0.1s)(s+10)}$$

Determine the static error coefficients K_p , K_v and K_a .

Solution :

$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

$$= \lim_{s \rightarrow 0} \frac{50}{(1+0.1s)(s+10)} = 5$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{50}{(1+0.1s)(s+10)} = 0$$

$$K_a = \lim_{s \rightarrow 0} s^2 \frac{50}{(1+0.1s)(s+10)} = 0$$

EXAMPLE 3.2. The forward path transfer function of a unity feedback control system given by

$$G(s) = \frac{5(s^2 + 2s + 100)}{s^2(s+5)(s^2 + 3s + 10)}$$

Determine the step, ramp and parabolic error coefficients. Also determine the type of the system.

Solution :

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{5(s^2 + 2s + 100)}{s^2(s+5)(s^2 + 3s + 10)}$$

$$K_p = \infty \quad \boxed{K_p = \infty}$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) = \lim_{s \rightarrow 0} s \cdot \frac{5(s^2 + 2s + 100)}{s^2(s+5)(s^2 + 3s + 10)}$$

$$K_v = \infty \quad \boxed{K_v = \infty}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \cdot \frac{5(s^2 + 2s + 100)}{s^2(s+5)(s^2 + 3s + 10)}$$

$$= 10 \quad \boxed{K_a = 10}$$

In denominator the value of $m = 2$. Hence, the given system is type '2' system.

EXAMPLE 3.3. The block diagram of an electronic pacemaker is given in Fig. 3.2. Determine the steady state error for unit ramp input when $K = 400$. Also, determine the value of K for which the steady state error to a unit ramp will be 0.02.

Solution : Given that $K = 400$

$$R(s) = \frac{1}{s^2}$$

$$H(s) = 1$$

$$\therefore G(s)H(s) = \frac{K}{s(s+20)}$$



Fig. 3.2.

$$\text{Steady state error is given by } e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{K}{s(s+20)}} = \lim_{s \rightarrow 0} \frac{s+20}{s(s+20)+400} = 0.05 \quad \text{Ans.}$$

Now $e_{ss} = 0.02$ (given)

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{K}{s(s+20)}} \\ 0.02 &= \lim_{s \rightarrow 0} \frac{s+20}{s(s+20)+K} \\ 0.02 &= \frac{20}{K} \quad \therefore \quad \boxed{K = 1000} \quad \text{Ans.} \end{aligned}$$

EXAMPLE 3.4. For a unity feedback control system the forward path transfer function is given by

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}$$

Determine the steady state error of the system. When the inputs are (i) 5 (ii) $5t$ (iii) $\frac{3t^2}{2}$.

Solution :

$$(i) \quad r(t) = 5 \quad \therefore \quad R(s) = \frac{5}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s} \cdot \frac{1}{1 + \frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s \rightarrow 0} \frac{5s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$\therefore e_{ss} = 0$$

$$(ii) \quad R(s) = \frac{5}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s^2} \cdot \frac{1}{1 + \frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s \rightarrow 0} s \cdot \frac{5}{s^2} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$e_{ss} = 10$$

$$(iii) \quad R(s) = \frac{3}{s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{3}{s^3} \cdot \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20}$$

$$e_{ss} = \infty$$

EXAMPLE 3.5. The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$$

Find the static error coefficients and steady state error of the system when subjected to an input given by

$$r(t) = 2 + 5t + 2t^2$$

Solution :

$$K_p = \lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s) = \lim_{s \rightarrow 0} s \cdot \frac{108}{s^2(s+4)(s^2+3s+12)} = \infty$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = \lim_{s \rightarrow 0} s^2 \cdot \frac{108}{s^2(s+4)(s^2+3s+12)} = \frac{108}{48}$$

$$r(t) = 2 + 5t + 2t^2$$

$$R(s) = \frac{2}{s} + \frac{5}{s^2} + \frac{4}{s^3}$$

$$\therefore e_{ss} = \frac{R_1}{1+K_p} + \frac{R_2}{K_v} + \frac{R_3}{K_a}$$

$$= \frac{2}{1+\infty} + \frac{5}{\infty} + \frac{4 \times 48}{108} = 1.77 \quad \boxed{e_{ss} = 1.77} \quad \text{Ans.}$$

EXAMPLE 3.6. Determine the type of the following unity feedback systems for which the forward path transfer function are given

$$(a) G(s) = \frac{K}{(s+10)(s+5)} \quad (b) G(s) = \frac{20}{s(1+0.1s)(1+0.05s)}$$

$$(c) G(s) = \frac{20(s+2)}{s^3(s+2)(s^2+5s+5)} \quad (d) G(s) = \frac{K}{s(s+1)(s+2)}$$

Solution : (a) Since in denominator the power of s is zero i.e. $m = 0$, hence it is type zero system.
 (b) type one system
 (c) $m = 3$, type '3' system
 (d) $m = 1$, type '1' system

EXAMPLE 3.7. A servomechanism is designed to keep a radar antenna pointed at a flying aeroplane. If the aeroplane is flying with a velocity of 600 km/hr, at a range of 2 km. and the maximum tracking error is to be within 0.1° , determine the required velocity error coefficient.
 (GATE, 1994)

Solution : The block diagram of servomechanism is shown in Fig. 3.3.

Linear velocity = 600 km./hr

$r = 2$ km.

Angular velocity $\omega = \frac{600}{2} = 300$ rad/hr.

$$= \frac{300}{3600} \text{ rad/sec.} = 0.083 \text{ rad/sec.}$$

$$R(s) = \frac{0.083}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)} = \lim_{s \rightarrow 0} s \cdot \frac{0.083}{s^2} \cdot \frac{1}{1 + \frac{K_v}{s(1+sT)}}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{0.083}{s} \cdot \frac{s(1+sT)}{s(1+sT) + K_v} = \frac{0.083}{K_v} = 0.1^\circ \text{ given}$$

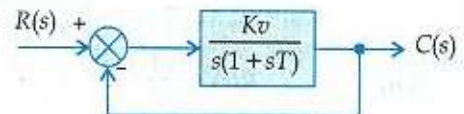


Fig. 3.3.

$$0.1 = \frac{0.083}{K_v}$$

$$\therefore K_v = 0.83 \text{ per degree}$$

$\therefore$ Required velocity error coefficient $K_v = 0.83$ per degree.

EXAMPLE 3.8. For the system shown in Fig. 3.4. Determine K_p and e_{ss} for unit step input.

Solution : $K_p = \lim_{s \rightarrow 0} G(s)H(s)$

$$= \lim_{s \rightarrow 0} \frac{1}{(s^2 + s + 2)(s + 1)} = \frac{1}{2}$$

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + 1/2} = \frac{1}{3/2} = \frac{2}{3}$$

$$K_p = 1/2$$

$$e_{ss} = 2/3$$

Ans.

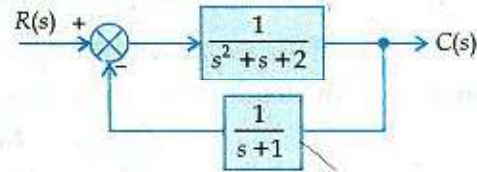


Fig. 3.4.

EXAMPLE 3.9. For a given system shown in Fig. 3.5, determine the actuating signal $E_a(s)$. Also determine the position error constant for unit step input.

Solution :

$$E_a(s) = 20 R(s) - B(s)$$

$$C(s) = \frac{20 E_a(s)}{s(1 + 0.05s)}$$

$$B(s) = C(s) \cdot \frac{1 + 0.01s}{1 + 0.05s}$$

$$B(s) = \frac{20(1 + 0.01s)E_a(s)}{s(1 + 0.05s)^2}$$

$$\therefore E_a(s) = 20R(s) - \frac{20(1 + 0.01s)E_a(s)}{s(1 + 0.05s)^2}$$

$$E_a(s) \left[1 + \frac{20(1 + 0.01s)}{s(1 + 0.05s)^2} \right] = 20 R(s)$$

$$\therefore E_a(s) = \frac{20R(s)}{1 + \frac{20(1 + 0.01s)}{s(1 + 0.05s)^2}}$$

$$E_a(s) = \frac{20s(1 + 0.05s)^2 \cdot R(s)}{s(1 + 0.05s)^2 + 20(1 + 0.01s)}$$

Ans.

For step input $R(s) = \frac{1}{s}$

$$E_a(s) = \frac{20(1 + 0.05s)^2}{s(1 + 0.05s)^2 + 20(1 + 0.01s)}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot \frac{20(1 + 0.05s)^2}{s(1 + 0.05s)^2 + 20(1 + 0.01s)} = 0$$

But $e_{ss} = \frac{1}{1 + K_p}$ or $0 = \frac{1}{1 + K_p}$

$$\therefore K_p = \infty$$

Ans.

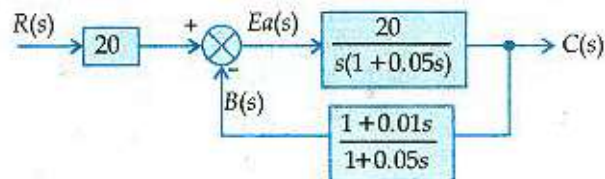


Fig. 3.5.

EXAMPLE 3.10. Consider a unity feedback control system with the closed loop transfer function

$$\frac{C(s)}{R(s)} = \frac{Ks+b}{s^2+as+b}$$

Determine the open loop transfer function. Show that the steady state error in the unit ramp input response is given by

$$e_{ss} = \frac{a-k}{b}$$

Solution : For closed loop control system

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$

$$\frac{Ks+b}{s^2+as+b} = \frac{G(s)}{1+G(s)}$$

$$\therefore G(s) = \frac{Ks+b}{s^2+s(a-K)}$$

$$\therefore \text{Open loop transfer function } G(s) = \frac{Ks+b}{s^2+s(a-K)}$$

Ans.

$$\text{Steady state error } e_{ss} = \lim_{s \rightarrow 0} s \cdot R(s) \cdot \frac{1}{1+G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{Ks+b}{s^2+s(a-K)}} \cdot 1 = \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s^2+s(a-K)}{s^2+s(a-K)+Ks+b}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s[s+a-K]}{s^2+s(a-K)+Ks+b} = \frac{a-K}{b}$$

$$e_{ss} = \frac{a-K}{b} \quad \text{Proved.}$$

3.5. DYNAMIC ERROR COEFFICIENTS

For the steady-state error, the static error coefficients gives the limited information.

The error function is given by

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)} \quad \dots(3.10)$$

For unity feedback system

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)} \quad \dots(3.11)$$

The equation (3.11) can be expressed in polynomial form (ascending power of 's')

$$\frac{E(s)}{R(s)} = \frac{1}{K_1} + \frac{1}{K_2}s + \frac{1}{K_3}s^2 + \dots \quad \dots(3.12)$$

CHAPTER 5

Stability Theory

Inside this Chapter

5.1. Concept of Stability; 5.2. Effect of Location of Poles on Stability; 5.3. Necessary But Not Sufficient Conditions for Stability; 5.4. The Routh-hurwitz Criterion; 5.5. Application of Routh's Stability Criterion to Control System Analysis; 5.6. Relative Stability Analysis; 5.7. Root Locus; 5.8. Rules for Construction of Root Loci; 5.9. Determination of K on Root Loci; 5.10. Effect of Addition of Poles and Zeros on Root Locus; 5.11. Introduction; 5.12. Mapping; 5.13. Mapping Of Close Contour and Principle of Argument; 5.14. Nyquist Path or Nyquist Contour; 5.15. Nyquist Criterion; 5.16. General Construction Rules of the Nyquist Path; 5.17. Analysis of Stability by Lyapunov's Direct Method; 5.18. Sylvester's Theorem; 5.19. How to Obtain Open Loop Transfer Function G(s) H(s) from Characteristic Equation

5.1. CONCEPT OF STABILITY

The concept of stability is very important to analyse and design the system. A system is said to be stable if its response cannot be made to increase indefinitely by the application of a bounded input excitation. If the output approaches towards infinite value for sufficiently large time, the system is said to be unstable.

A linear time invariant (LTI) system is stable if

1. The system is excited by a bounded input, the output is bounded (BIBO stability criteria).
2. In the absence of the input, the output tends towards zero (the equilibrium state of the system).

This is known as asymptotic stable.

Consider the transfer function

$$\frac{C(s)}{R(s)} = \frac{a_m s^m + a_{m-1} s^{m-1} + \dots + a_1 s + a_0}{b_n s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0} \quad \dots(5.1)$$

The output is given by

$$C(t) = \int_0^{\infty} g(\tau) r(t - \tau) d\tau \quad \dots(5.2)$$

where $g(\tau) = \mathcal{L}^{-1} G(s)$ = impulse response of the system. So, a system is said to be stable if the impulse response approaches zero for sufficiently large time. If the impulse response approaches infinity for sufficiently large time, the system is said to be unstable. If the impulse response approaches a constant value for sufficiently large time, the system is said to be marginally stable.

5.2. EFFECT OF LOCATION OF POLES ON STABILITY

(a) Poles on Negative Real Axis

Consider a simple pole at $s = -a$ as shown in fig. 5.1(a), the corresponding impulse response is given by

$$g(t) = \mathcal{L}^{-1} G(s) = \mathcal{L}^{-1} \frac{K}{s+a} = K.e^{-at} \quad \dots(5.3)$$

As the time 't' increases, the response approaches zero and the system is stable. The response is shown in fig 5.1(b).



Fig. 5.1.

(b) Pole on Positive Real Axis

Consider a system having simple pole on positive real axis at $s = a$, the corresponding impulse response is given by

$$g(t) = \mathcal{L}^{-1} \frac{K}{s-a} = K.e^{at} \quad \dots(5.4)$$

The response increases exponentially with time, hence the system is unstable. The simple pole and response are shown in Fig. 5.2 (a) and (b).

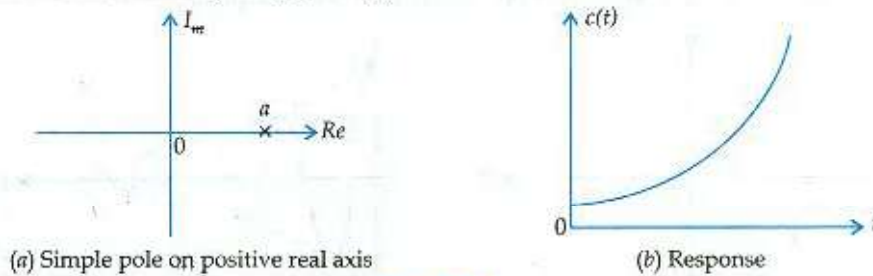


Fig. 5.2.

(c) Pole at the Origin : Consider a Pole at Origin

$$\therefore g(t) = \mathcal{L}^{-1} \frac{K}{s} = K \quad \dots(5.5)$$

This is constant value, hence the system is marginally stable. If there are two poles at the origin, the time response would be

$$g(t) = \mathcal{L}^{-1} \frac{K}{s^2} = Kt \quad \dots(5.6)$$

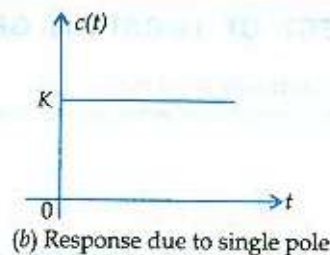
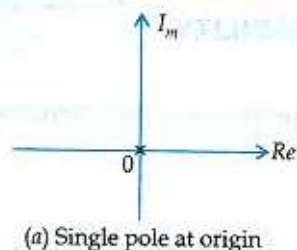


Fig. 5.3.

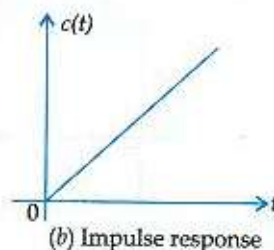
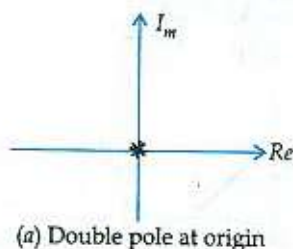


Fig. 5.4.

(d) Complex Pole in the Left Half of s-plane

Let the transfer function has a complex conjugate poles at $s = -\alpha \pm j\omega$. The time response due to the complex conjugate poles is given by

$$g(t) = \mathcal{L}^{-1} \left[\frac{K}{s + \alpha - j\omega} + \frac{K}{s + \alpha + j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2K(s + \alpha)}{(s + \alpha)^2 + \omega^2} \right] = 2K e^{-\alpha t} \cos \omega t \quad \dots(5.7)$$

When t increases $g(t)$ approaches zero and the system is stable. The complex poles and corresponding time response is shown in Fig. 5.5(a) and 5.5(b) respectively.

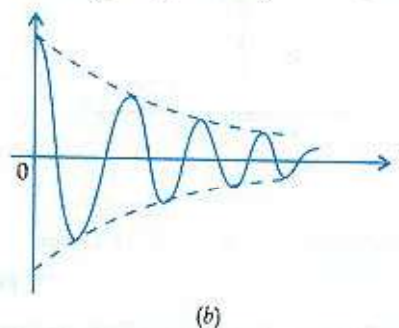
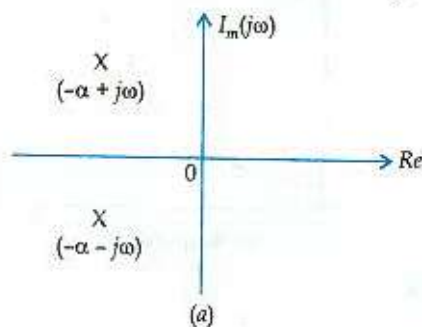


Fig. 5.5.

(e) Complex Poles in the Right Half of s-plane

Suppose the system has complex conjugate poles at $s = \alpha \pm j\omega$. The time response is given by

$$g(t) = \mathcal{L}^{-1} \left[\frac{A}{s - \alpha - j\omega} + \frac{A}{s - \alpha + j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2A(s - \alpha)}{(s - \alpha)^2 + \omega^2} \right] = 2A e^{\alpha t} \cos \omega t \quad \dots(5.8)$$

Hence, the response increases exponentially sinusoid with time and therefore the response is unstable. The poles and time response shown in Fig. 5.6(a) and 5.6(b) respectively.

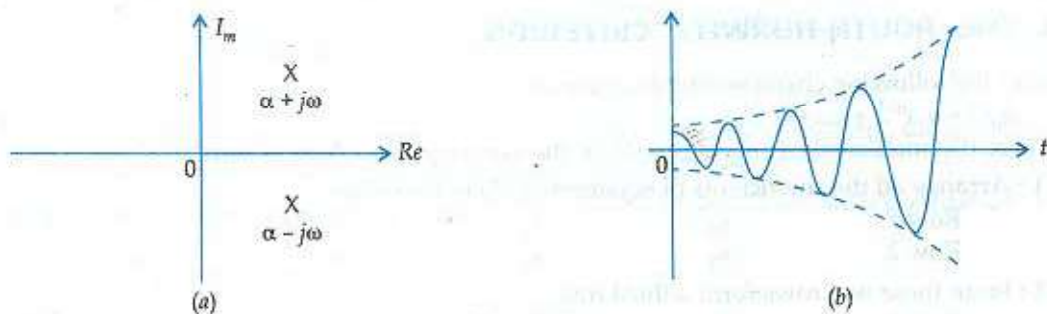


Fig. 5.6.

(f) Poles on $j\omega$ -axis

If the system having the complex poles on $j\omega$ -axis the corresponding time response would be

$$g(t) = \mathcal{L}^{-1} \left[\frac{A}{s + j\omega} + \frac{A}{s - j\omega} \right] = \mathcal{L}^{-1} \left[\frac{2As}{s^2 + \omega^2} \right] = 2A \cos \omega t \quad \dots(5.9)$$

The response is marginally stable. The equation (5.9) shows the sustained oscillations of constant amplitude. This situation will also be considered unstable.

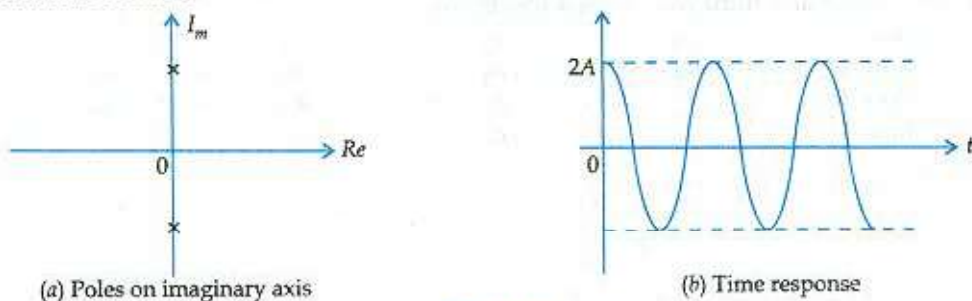


Fig. 5.7.

The overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad \dots(5.10)$$

$$\text{The characteristic equation is } 1 + G(s)H(s) = 0. \quad \dots(5.11)$$

The necessary and sufficient condition that a feedback system be stable is that all the zeros of the characteristic equation $1 + G(s)H(s) = 0$ have negative real part. Or, in terms of poles we can say that the necessary and sufficient condition that a feedback system be stable is that all the poles of overall transfer function have negative real part.

5.3. NECESSARY BUT NOT SUFFICIENT CONDITIONS FOR STABILITY

Consider a system with characteristic equation

$$a_0 s^m + a_1 s^{m-1} + \dots + a_m = 0 \quad \dots(5.12)$$

- (a) All the coefficients of the equation should have same sign,
- (b) There should be no missing term.

If above two conditions are not satisfied the system will be unstable. But if all the coefficients have same sign and there is no missing term we have no guarantee that the system will be stable. For stability we use Routh-Hurwitz Criterion.

5.4. THE ROUTH-HURWITZ CRITERION

Consider the following characteristic polynomial

$$a_0 s^n + a_1 s^{n-1} + \dots + a_n = 0 \quad \dots(5.13)$$

where the coefficients $a_0, a_1, \dots, a_n$ are all of the same sign and none is zero.

Step 1 : Arrange all the coefficients of equation (5.13) in two rows

Row 1	a_0	a_2	a_4	
Row 2	a_1	a_3	a_5	

Step 2 : From these two rows form a third row

Row 1	a_0	a_2	a_4	
Row 2	a_1	a_3	a_5	
Row 3	b_1	b_3	b_5	

where,

$$b_1 = -\frac{1}{a_1} \begin{vmatrix} a_0 & a_2 \\ a_1 & a_3 \end{vmatrix}$$

$$b_3 = -\frac{1}{a_1} \begin{vmatrix} a_0 & a_4 \\ a_1 & a_5 \end{vmatrix}$$

Step 3 : From second and third row, form a fourth row

Row 1	a_0	a_2	a_4	
Row 2	a_1	a_3	a_5	
Row 3	b_1	b_3	b_5	
Row 4	c_1	c_3	c_5	

where,

$$c_1 = -\frac{1}{b_1} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}$$

$$c_3 = -\frac{1}{b_1} \begin{vmatrix} a_1 & a_5 \\ b_1 & b_5 \end{vmatrix}$$

:

:

Step 4 : Continue this procedure of forming a new rows.

5.4.1. Statement of Routh-Hurwitz Criterion

Routh-Hurwitz criterion states that the system is stable if and only if all the elements in the first column have the same algebraic sign. If all elements are not of the same sign then the number of sign changes of the elements in first column equals the number of roots of the characteristic equation in the right half of the s-plane (or equals to the number of roots with positive real parts).

EXAMPLE 5.1. Check the stability of the system whose characteristic equation is given by

$$s^4 + 2s^3 + 6s^2 + 4s + 1 = 0$$

Solution : Obtain the array of coefficients as follows

s^4	1	6	1
s^3	2	4	
s^2	4	1	
s^1	3.5		
s^0	1		

$$b_1 = -\frac{1}{2} \begin{vmatrix} 1 & 6 \\ 2 & 4 \end{vmatrix} = 4, \quad c_1 = -\frac{1}{4} \begin{vmatrix} 2 & 4 \\ 4 & 1 \end{vmatrix} = 3.5$$

EXAMPLE 5.5. A closed loop control system has the characteristic equation given by

$$s^3 + 4.5s^2 + 3.5s + 1.5 = 0$$

Investigate the stability using Routh-Hurwitz criterion.

(R.M.L. University Faizabad, 2001)

Solution :

s^3	1	3.5
s^2	4.5	1.5
s^1	3.17	
s^0	1.5	

No. of sign changes in first column = 0

No. of roots in right half of s -plane = 0

Hence, system is stable.

EXAMPLE 5.6. Check the stability of the system, having following characteristic equation.

Solution: $s^5 + 6s^4 + 3s^3 + 2s^2 + s + 1 = 0$

s^5	1	3	1
s^4	6	2	1
s^3	2.67	0.83	
s^2	0.135	1	
s^1	-18.95		
s^0	1		

No. of sign change in first column = 2

No. of poles on right half of s -plane = 2

Hence, system is unstable.

SPECIAL CASES

Case 1 : If a first column term in any row is zero, but the remaining terms are not zero or there is no remaining term, then multiply the original equation by a factor $(s + a)$ where ' a ' is any positive real number. The simplest value of ' a ' is 1 (take $a = 1$). Consider the following example.

EXAMPLE 5.7. Investigate the stability

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Solution :

s^5	1	2	3
s^4	1	2	5
s^3	0		
s^2			
s^1			
s^0			

Now, multiply the characteristic equation by $(s + 1)$

$$(s + 1)(s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5) = 0$$

$$\text{or, } s^6 + 2s^5 + 3s^4 + 4s^3 + 5s^2 + 8s + 5 = 0$$

s^6	1	3	5	5
s^5	2	4	8	
s^4	1	1	5	
s^3	2	-2		
s^2	2	5		
s^1	-7			
s^0	1			

From the above table :

No. of sign change in the first column = 2

No. of roots in the right half of s -plane = 2

Hence, system is unstable.

EXAMPLE 5.8. Apply Routh-Hurwitz criterion to the following equation and investigate the stability.

$$s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0$$

Solution :

s^5	1	2	11
s^4	2	4	10
s^3	0		
s^2			
s^1			
s^0			

Since, the third element in first column is zero, so multiply the equation by $(s + 1)$

$$(s + 1)(s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10) = 0$$

or $s^6 + 3s^5 + 4s^4 + 6s^3 + 15s^2 + 21s + 10 = 0$

s^6	1	4	15	10
s^5	3	6	21	
s^4	2	8	10	
s^3	-6	6		
s^2	10	10		
s^1	12			
s^0	1			

From the above table it is clear that there are two changes of sign in first column, therefore there are two roots in the right half of s -plane. The system is unstable.

ALTERNATIVE METHOD

Replace the zero by a small positive quantity ϵ and continue the procedure. Consider the following example.

EXAMPLE 5.9. Consider the following equation

$$s^3 + s + 2 = 0$$

Solution :

s^3	1	1
s^2	0	2
s^1		
s^0		

Replace 0 by ϵ and continue the procedure.

s^3	1	1
s^2	ϵ	2
s^1	$-\frac{1}{\epsilon}(2 - \epsilon)$	-
s^0	2	

No. of sign changes in first column = 2

No. of roots in right of s -plane = 2

Case 2 : When any one row of Routh table is zero.

When any one row of Routh table is zero, it indicates that the equation has at least one pair of roots which lie radially opposite each other and equidistant from origin. The array can be completed by forming the auxiliary polynomial. The polynomial whose coefficients are the element of the row just above the row of zeros in Routh array is called an auxiliary polynomial. Consider the following example

EXAMPLE 5.10. Consider the equation

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

Solution :

s^5	1	24	-25
s^4	2	48	-50
s^3	0	0	
s^2			
s^1			
s^0			

From above table it is clear that the third row is zero. Hence form the auxiliary polynomial. $A(s)$

$$A(s) = 2s^4 + 48s^2 - 50$$

$$\frac{dA(s)}{ds} = 8s^3 + 96s$$

Now the Routh array can be written as

s^5	1	24	-25
s^4	2	48	-50
s^3	8	96	
s^2	24	-50	
s^1	112.6		
s^0	-50		

No. of sign changes in first column = 1

No. of roots in right half of s -plane = 1

The roots of the equation formed by the auxiliary polynomial

$$2s^4 + 48s^2 - 50 = 0$$

are also the roots of the original equation.

The given equation can be written in factored form as

$$(s+1)(s-1)(s+j5)(s-j5)(s+2) = 0$$

It is clear that the original equation has one root with positive real part.

The roots of the auxiliary equation are also the roots of the characteristic equation because auxiliary equation is the part of characteristic equation. Suppose, original characteristic equation is of order six *i.e.*, having six roots and auxiliary equation is of order four *i.e.*, having four roots. These four roots are also the roots of the original characteristic equation and remaining $(6-4=2)$ two roots will always lie to the left half of s -plane and these two roots do not have significant role for stability because they lie far away from imaginary axis. The roots of the auxiliary equations are dominant roots,* the stability can also be determined from the roots of the auxiliary equation.

Consider the following characteristic equation :

$$S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$$

S^6	1	8	20	16
S^5	2	12	16	0
S^4	2	12	16	0
S^3	0	0	0	0
S^2				
S^1				
S^0				

Auxiliary equation $A(s) = 2S^4 + 12S^2 + 16 = 0$, $\frac{dA(s)}{ds} = 8S^3 + 24S$. Since, the order of the characteristic equation is six and order of auxiliary equation is four, the two roots $(6-4=2)$ lie to the left half of s -plane and four roots of auxiliary equations are dominant roots.

* Dominant roots means the roots are close to the imaginary axis or on the imaginary axis or they are in right half of s -plane.

S^6	1	8	20	16
S^5	2	12	16	0
S^4	2	12	16	0
S^3	8	24	0	0
S^2	6	16	0	
S^1	2.67	0		
S^0	16			

No sign change in first column, the system may be stable, since there is a row of zeros it means the roots are lying on imaginary axis. So solve the auxiliary equation $A(s) = 0$.

$$2S^4 + 12S^2 + 16 = 0$$

Put $S^2 = x$

$$\therefore 2x^2 + 12x + 16 = 0$$

$$\text{or } x^2 + 6x + 8 = 0$$

$$(x+2)(x+4) = 0$$

$$x+2=0, \quad x+4=0$$

$$x = -2, \quad x = -4$$

$$S^2 = -2, \quad S^2 = -4$$

$$S = \pm j\sqrt{2}, \quad S = \pm j2$$

Since, the roots are nonrepeated on imaginary axis, hence the system is marginally stable.

No. of roots with positive real part = 0

No. of roots with negative real part = 2

No. of roots with zero real part = 4

Important Note : If Sign changes in first column, system is unstable. The number of roots with positive real part of characteristics equation will be equal to the number of sign changes i.e., the roots are located in right half of s-plane. But if no sign changes in first column the system may be stable. In such case stability can be determined by solving auxiliary equation for roots and from the location of roots, stability can be determined.

5.5. APPLICATION OF ROUTH'S STABILITY CRITERION TO CONTROL SYSTEM ANALYSIS

Routh stability criterion is also used for the determination of stability of the linear feedback systems. Consider the following example.

EXAMPLE 5.11. The open loop transfer function of unity feedback system is

$$\frac{K}{s(1+0.4s)(1+0.25s)}$$

Find the restriction of K so that the closed loop system is absolutely stable.

(R.M.L. University Faizabad, 2001)

Solution : Given that

$$G(s) = \frac{K}{s(1+0.4s)(1+0.25s)}$$

$$H(s) = 1$$

The characteristic equation $1 + G(s)H(s) = 0$

$$1 + \frac{K}{s(1+0.4s)(1+0.25s)} = 0$$

$$\text{or } s(1+0.4s)(1+0.25s) + K = 0$$

$$\text{or } s^3 + 6.5s^2 + 10s + 10K = 0$$

s^3	1	10
s^2	6.5	10K
s^1	$\frac{65 - 10K}{6.5}$	
s^0	10K	

For absolute stability, there should be no sign change in the first column i.e., no root of the characteristic equation should lie in right half of s-plane. This is possible only when

$$K > 0 \text{ and } 65 - 10K > 0$$

Hence, for closed loop stability

$$0 < K < 6.5$$

5.6. RELATIVE STABILITY ANALYSIS

Routh stability criterion gives the information about the absolute stability. But we are more interested for the relative stability of the system. Relative stability can be examined by shifting the s-plane and then apply Routh stability criterion. The characteristic equation is modified by shifting the origin of s-plane to $s_1 = -\sigma$ by the substitution

$$s = z - s_1$$

Now apply the Routh stability criterion, the number of sign change in the first column is equal to the number of roots to the right of the vertical line $s_1 = -\sigma$ or in other words the roots of the original characteristic equation are more negative than $-\sigma_1$.

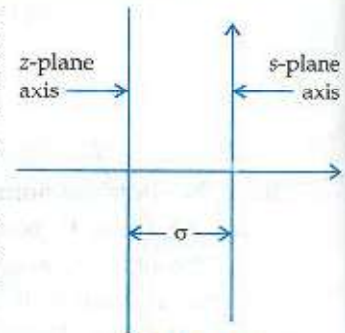


Fig. 5.8.

EXAMPLE 5.12. Check whether all the roots of the equation $s^3 + 7s^2 + 25s + 39 = 0$ have real parts more negative than -1 .

Solution :

s^3	1	25
s^2	7	39
s^1	19.42	
s^0	39	

Since, there is no sign change in the first column, it means all the roots are in the left half of s-plane.

Put $s = z - 1$ in the characteristic equation

$$(z-1)^3 + 7(z-1)^2 + 25(z-1) + 39 = 0$$

$$\text{or, } z^3 + 4z^2 + 14z + 20 = 0$$

z^3	1	14
z^2	4	20
z^1	9	
z^0	20	

Since, no sign change in the first column, the roots of the characteristic equation lie in the left of z-plane, it means all the roots of the original equation in s-domain lie to the left of $s = -1$.

ILLUSTRATIVE EXAMPLES

EXAMPLE 5.13. The characteristic equation of a servosystem is given by

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$$

Determine the conditions which must be satisfied by the coefficient of the characteristic equation for the system to be stable.